

Carbon Data Is Becoming Economic Currency: Is Asia Ready?

Partnership for Carbon Transparency (PACT), WBCSD
April 2026



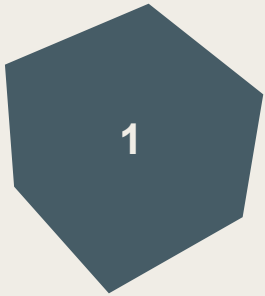
Who We Are

The leading global business community
turning sustainability into a source of
performance and competitive advantage.

230 global corporations: 57 headquartered in Asia Pacific



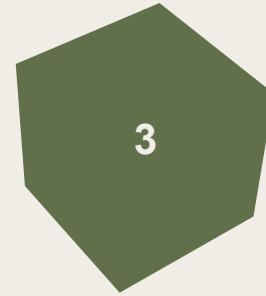
Across China, India, Japan and South East Asia, WBCSD is supporting its global member community to:



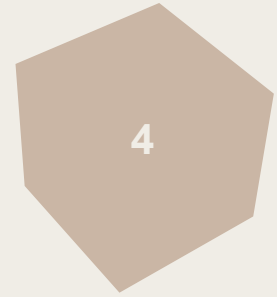
Accelerate emission reduction, meeting energy needs and unlocking business value



Strengthen the performance and accountability system to reward sustainable businesses



Build resilience and competitiveness along value chains



Upskill their boards and teams to thrive through the transition

New WBCSD resources out this week



Enabling investments that are aligned with transition ambition



A new APAC-focused insights extending the main report



An interview on Global vs. APAC physical risk trends



Accessed through "Download Publications" on the main report page.

Carbon used to be free.

Now it has a price.

And almost no one, not the buyer, not the seller, not the regulator, not the bank, can tell you, with any reliability, what the carbon content of a product actually is.

Powered by

Something is fundamentally changing.

Carbon used to be invisible. Now it has a price, and that price is moving supply chains.

REGULATION

Carbon borders are real

CBAM is live. EU importers now face a direct financial liability linked to the embedded emissions of traded goods. Other markets are watching, and following.

PROCUREMENT

Buyers are demanding it

Global buyers are shifting from "sustainability courtesy" to sourcing criterion. Product carbon data is becoming a condition of market access.

FINANCE

Capital is pricing it in

Banks, insurers, and investors are integrating product-level carbon performance into lending, risk pricing, and transition finance decisions.

*Carbon is becoming the **third variable of trade**, after price and quality.*

APAC is converging at pace on product-level carbon data

APAC is building the standards, infrastructure, and political will.



Policy signals are aligning

APAC governments are moving toward **product-level carbon rules**. Japan, Korea, China, India are each formalizing **PCF-based frameworks**.



Multinationals want one system

Heavy emitters in international supply chains (e.g. steel, chemicals, cement) are asking for **consistent PCF rules** to meet CBAM/DPP and global buyer expectations.



Building blocks exist

National platforms are emerging across the region (databases, digital MRV, GX Digital, TGO, Trusted Data Spaces).



Momentum for collaboration is growing

Regional alliances (SBF, CAII, JEITA, K-ETS partners) are actively exploring cross-border PCF cooperation.

But the data doesn't flow

The infrastructure to move product carbon data, reliably, at scale, across borders, doesn't yet exist.

~90%

Modelled or
approximated

Across incompatible
platforms

Siloed

~9%

of companies report
Scope 3
comprehensively

1973

Banks couldn't move money across borders without telegrams and telex operators.

Slow. Error-prone. Incompatible.

239 banks, 15 countries, built a way to talk to each other. They called it **SWIFT**.

Didn't replace banks.

Didn't standardize money.

Standardized the messages.

2026

PACT is the SWIFT of product-level carbon data

Not a platform. Not a database. Not another standard.

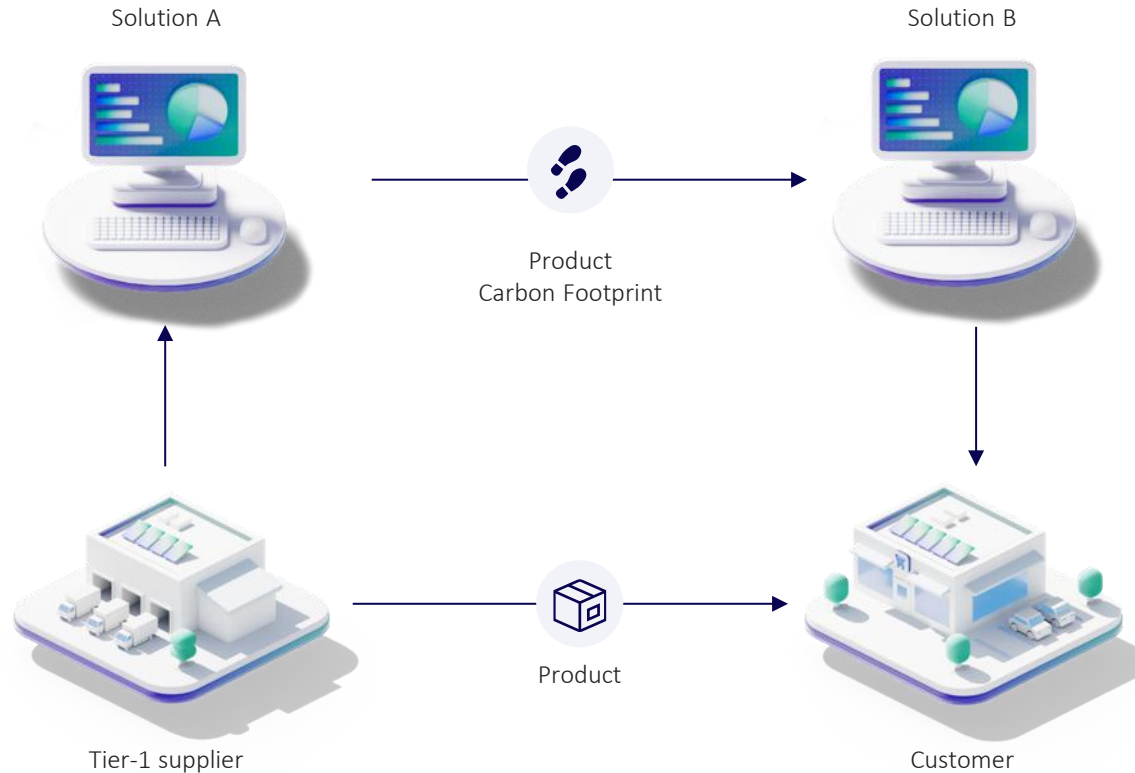
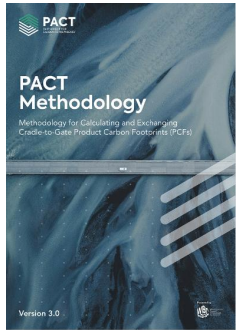
The connective tissue that lets carbon data flow across organizations, across value chains, and across borders.

97 multinationals, 50 solution providers, 11 industry-specific initiatives, coming together to build the carbon infrastructure we all need.

Measure once. Flow everywhere. Trusted by all parties.

PACT enables companies to unlock product-level carbon via three critical building blocs

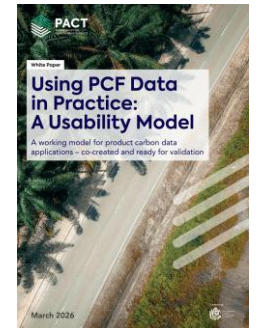
1. Globally applicable, harmonized product level carbon accounting rules across industries



2. A technical infrastructure enabling automated data exchange across interoperable solutions



3. Fit-for-purpose model unlocking use cases for product carbon data



Value chain collaboration to align on standards, drive adoption across suppliers and industries, and create trust. 150+ stakeholders and industry initiatives representing 5,000 companies involved

The diagram illustrates the ecosystem of stakeholders influencing the development of the Greenhouse Gas (GHG) Reporting Standards. The central circle features a stylized 'G' logo. Surrounding it are five segments, each representing a stakeholder group, with numerous logos of organizations belonging to that group.

- Corporate Members & Contributors:** Includes logos for ABInBev, ABB, Ahold Delhaize, Ajinomoto, Alibaba.com, Amazon, Apple, ARCADIS, APP, Avery Dennison, Ayala, BASF, ALTRUISTIQ, Asuene, atma.io, boost, carboledger, carbon trail, CircularTree, CEDARS, CIRRUS NEXUS, climate camp, carbmee, CO2 AI, DASSAULT SYSTEMES, DELTA, DIAGEO, ENGIE, dsm-firmenich, Roche, FUJITSU, Givaudan, HITACHI, Infineum, JM, JSW, KAO, KOMATSU, LYB, HowGood, InnoBlock, Microsoft, Makersite, OPTEL, PLANET→FWD, RISKOA, SAP, SIEMENS, SM, RT, TR, DZT, STARTRUST, SUSTECH, TERRALYTIQ, Terrascope, TRADE-VAN, TOPPAN, CTWS, uhuru, variable, YUZEDATA, Wm, zeroboard, ZeroTwentyFifty.
- PACT Conformant Solutions:** Includes logos for Bayer, bp, Chevron Phillips, CISCO, CLIMEWORKS, Coca-Cola, COLGATE-PALMOLIVE, Continental, CRI, DAIKIN, Danfoss, edp, FITSOL, FUJITSU, GreenStitch, HITACHI, InnoBlock, Microsoft, Makersite, OPTEL, PLANET→FWD, RISKOA, SAP, SIEMENS, SM, RT, TR, DZT, STARTRUST, SUSTECH, TERRALYTIQ, Terrascope, TRADE-VAN, TOPPAN, CTWS, uhuru, variable, YUZEDATA, Wm, zeroboard, ZeroTwentyFifty.
- Industry focused initiatives:** Includes logos for Catena-X, European Aluminium, GLOBAL BATTERY ALLIANCE, gfli, Green x Digital, INTERNATIONAL ALUMINIUM, International Copper Association, IEC, PLATINUM GROUP METALS, OPEN FOOTPRINT, GLEIF, ZINC, International ZINC association, NICKEL INSTITUTE, RMi, Smart Freight Centre, ValueChange, PSCI, WRAP, TOGETHER FOR SUSTAINABILITY.
- Standard Setters & Disclosure organizations:** Includes logos for Greenhouse Gas Protocol, ISO, WORLD RESOURCES INSTITUTE, SCIENCE BASED TARGETS, IFRS, CDP, ecovadis, GSI.
- Public Private Partnerships:** Includes logos for SME Climate Hub, SBF, THE CLIMATE PLEDGE 10 years Early, WORLD ECONOMIC FORUM, SCOPE 3.
- Auditors, Consultancies, Financial Institutions & Data Providers:** Includes logos for ACCOUNTANCY EUROPE, KPMG, Deloitte, EY, pwc, TÜVRheinland, BCG, Chiltern TMC, CARBON TRUST, SE ADVISORY SERVICES, Nexio Projects, NRI, WAP, AGECO, accenture, Guidehouse, BAIN & COMPANY, PRÉ, Carbon Accounting Alliance, ADB, citibank, ecoinvent, sphera.

We are not a pilot. We are running.

M E T H O D O L O G Y



~50

software solutions conformant with PACT

180

corporates exchanging data

5,000+

suppliers connected

4M+

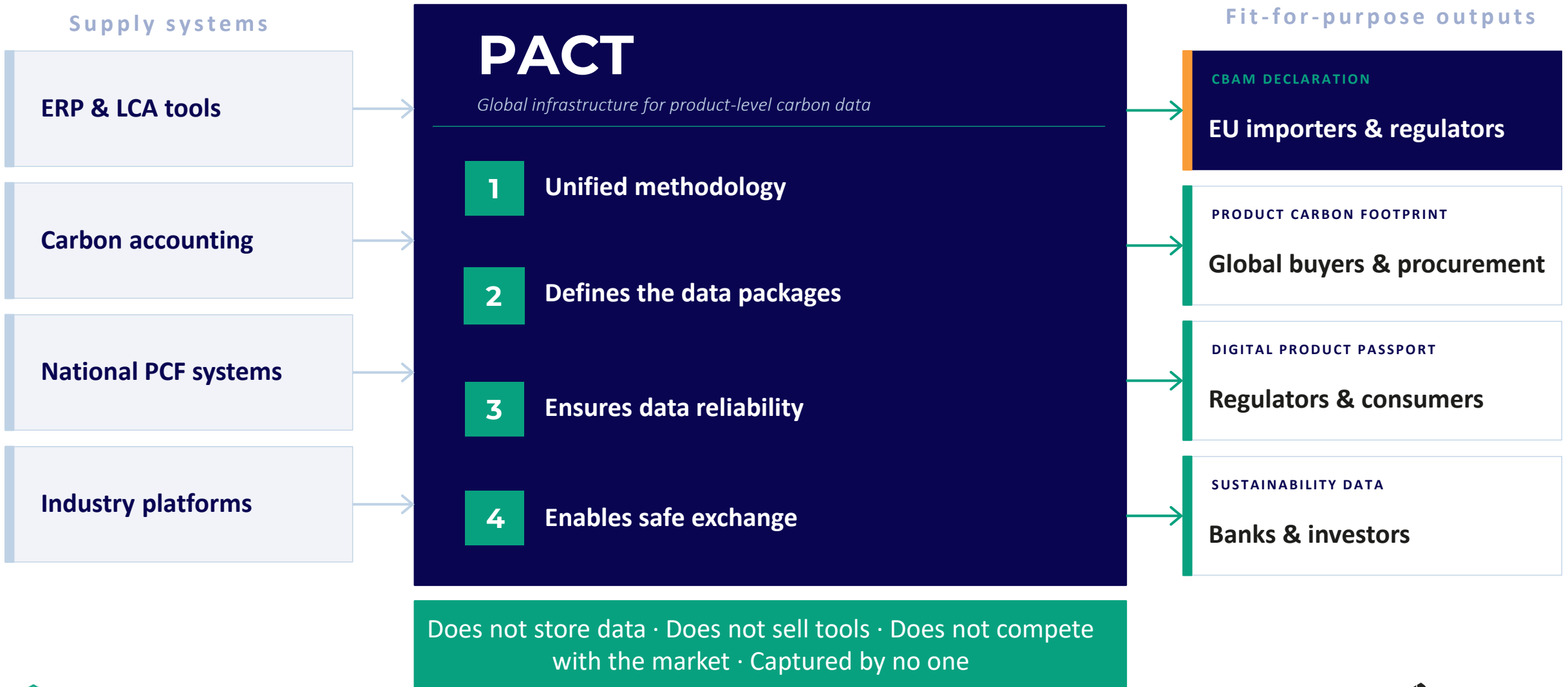
PCFs calculated in 2025

The largest network for product carbon footprint exchange in the world.

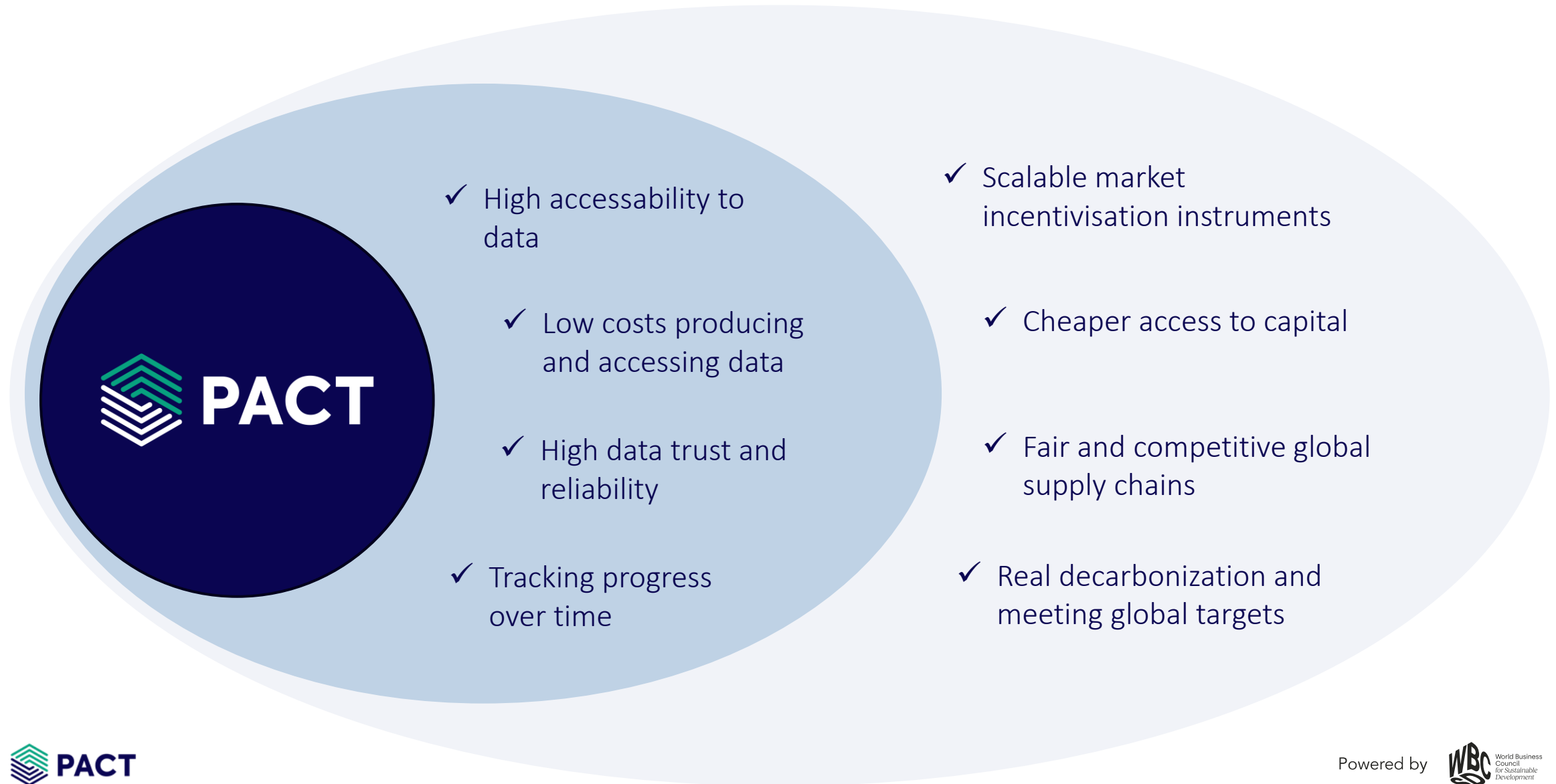
And still, a drop in the ocean.

The utility the market needs.

Global open infrastructure for product-level carbon data.



What becomes possible when product carbon data flows globally



APAC 2035: unlocking the low-carbon industrial transition

Without trusted product-level carbon data

- ✗ APAC exporters face rising trade friction
- ✗ High-cost penalties hit key sectors
- ✗ Capital and procurement shift away from APAC
- ✗ High-carbon lock-in accelerates

APAC loses competitiveness. The world loses the climate battle.

2035

With trusted product-level carbon data

- ✓ APAC suppliers stay competitive in global markets
- ✓ Carbon becomes a competitive advantage
- ✓ Capital flows into APAC industrial upgrading
- ✓ Decarbonization accelerates across industries

APAC becomes a leading engine of low-carbon global trade.

Today's speakers



Naama Avni-Kadosh

Director, Partnership for
Carbon Transparency,
WBCSD



Samantha Woods

Director, Sustainability,
Swire Coca Cola



Oswald Kuyler

Senior Digital Trade Advisor,
Asian Development Bank
(ADB)



Yiran Guo

Director, Policy, National
Climate Change Secretariat
Singapore

We will now hear from..

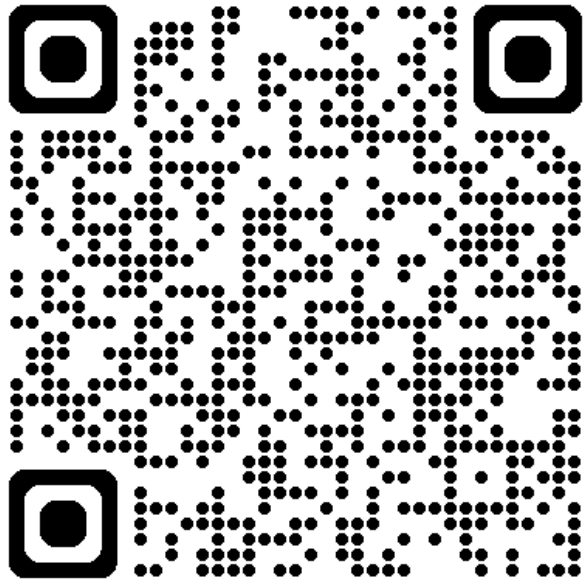


Hu Ching

Head, Net Zero
Transition
Programme Office,
Singapore Business
Federation



Your pocket guide to *PACT*



*Download the
PACT Cheat Sheet
Everything your team
needs, in one place*

Turn your sustainability data into a business asset with PACT

Use PACT Infrastructure

- Adopt PACT methodology / PACT aligned industry specific methodology
- WBCSD Implementation program
- PACT conformant IT solution
- PACT aligned consultancy

Design PACT Infrastructure

- Join the CBAM pilots and working groups
- Test PACT Usability Model
- Design next data packages in PACTs Use case forum
- Pilot data exchange with PACT Sand Box environment

Contact us for resources, support, and inquiries: pact@wbcsd.org

Building the Carbon Data Network: From Interoperability to Implementation

Partnership for Carbon Transparency (PACT), WBCSD
May 2026



**PACT is building the trusted global
infrastructure for product carbon data,
powering competitive and resilient supply
chains**

What decision-grade carbon data actually requires.

The market is setting the bar. PACT is the infrastructure that meets it.

01

Disclose once, reuse many

One calculation. Multiple outputs. Buyers, regulators, voluntary schemes, without repeating the work.

02

Confidential but defensible

Sensitive data stays protected. Disclosed results still stand up to audit and regulatory scrutiny.

03

Comparable + decision-grade

Not just a number, a number you can act on. Benchmarks and quality signals that drive real investment decisions.

04

Ecosystem enablement

Common rules for every system, sector and verifier. Break silos. Scale the full value chain.

Three layers. One infrastructure

01

Common methodological base

A shared foundation ensuring product carbon footprints are calculated consistently and are comparable across sectors and borders.

02

Technical exchange standard

An interoperability protocol so data flows between any software, any system, any jurisdiction without manual conversion or reformatting.

03

Fit-for-purpose model

A quality-assurance layer that tells recipients exactly what decisions each data point can credibly support from directional prioritization to regulatory filings.

The PACT Methodology is PACT's industry-agnostic emissions accounting methodology, which was launched in April 2025 (v3)

Context

Why?

Harmonization: diverse and diverging approaches to product-level accounting

Consistency: too much flexibility in current accounting methodologies

Ambition: Encouraging companies to get started and providing pathways to improve



Building blocks



Emissions accounting

- Existing methods and standards
- Scope and boundary
- Calculating PCFs



Data integrity

- Data sources and hierarchy
- Primary data share calculation
- Data quality assessment



Verification

- Verification roadmap



Data Exchange

- Data attributes for data exchange
- Integrating PCFs into Scope 3 inventories

The PACT Technical Specifications enable the interoperable exchange of Product Carbon Footprints across PACT Conformant Solutions

PACT Technical Specifications V3

Technical Specifications for PCF Data Exchange (3.0.0-20250428)

Living Document

Latest published version:

<https://docs.carbon-transparency.org/spec/3.0.0/>

Previous Versions:

<https://docs.carbon-transparency.org/spec/2.3.1/>

Feedback:

pact@wbcsd.org with subject line "[data-exchange-protocol] ... message topic ..."

[GitHub](#)

Abstract

This document specifies a data model for GHG emission data at product level based on the PACT Methodology (previously Pathfinder Framework) Version 3, and a protocol for interoperable exchange of GHG emission data at product level.

§ 1. Introduction

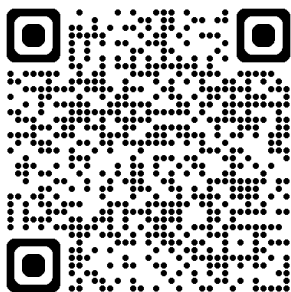
This document is a work in progress and should not be used for conformance testing. Please refer to the latest stable version of the Technical Specifications for this.

For an overview of changes since the last version (2.3), see the [Appendix A: Changelog](#).

This document contains the necessary technical foundation for the [PACT Network](#), an open : for emission data exchange.

The goal of this document is to enable the [interoperable](#) exchange of [Product Carbon Footprint](#) ing host systems.

The methodological foundation of the specification is the PACT Methodology Version 3.0, se [\[PACT-METHODOLOGY\]](#).



Building blocks



Data Model

The data model specifies:

- Set of attributes, attribute definitions, and syntax
- Data model available as an open API schema



API Specification

Specifies a standard technical language for software solutions to send, request, and receive PCF information over the internet

From one-size-fits-all PCFs to tiered, fit-for-purpose PCFs

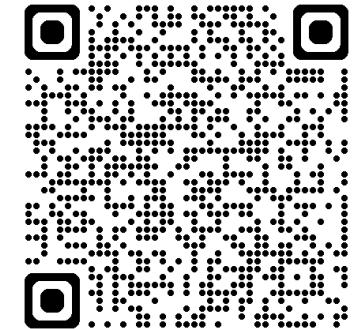
To scale product carbon data across global supply chains, data must be **usable** across different business and regulatory contexts.

Level	What the received PCF is (high level description)	What the received PCF shows (Data Attribute)	Where the received PCF has limitations (high level description)
Level 1: Indicative PCF	A minimal PCF dataset containing core identification, unit definition and cradle-to-gate emissions values	- Product name, description, relevant product IDs and classification codes, status - PCF excluding biogenic CO₂ uptake - Declared unit, unit and amount, and product mass per Declared unit - Reference period - Geographic Scope	- Majority of PACT SHALL attributes not populated - Data reliability not assessable - Sector-specific or optional contextual attributes likely absent
Level 2: Consistent PCF	A Level 1 dataset plus additional metadata enabling some evaluation of, data reliability, representativeness and methodological consistency	- All Level 1 attributes - PCF including biogenic CO₂ uptake - Biogenic carbon content, if applicable* - Primary Data Share (PDS) - Secondary emission factor sources - IPCC Characterisation Factors - Relevant standards used (product- or sector-specific rules), if applicable**	- Incomplete coverage of PACT SHALL attributes - Some methodological assumptions not fully documented - Limited methodological transparency
Level 3: Methodologically Complete PCF	A methodologically complete PCF with the majority of PACT data attributes populated, clear transparency on data sources, choices and exclusions, and data reliability indicators, demonstrating greater integrity than lower Levels	- All Level 2 attributes - Description of boundary processes 100% PACT SHALL attributes - Allocation rules - If exemption rules percentage >3%, exemption rules description	Current climate of PCF landscape means that it is likely that emission factors remain secondary data
Level +: Conditional PCF	A PCF at any Level where the PCF alone is insufficient to satisfy the application or decision. Additional use-case-specific alignment conditions and supplementary metadata must also be met, for example, to support a formal comparison, commercial decision, certification or external claim. The specific conditions required are defined by the relevant use case and will be developed through the PACT Product-level Data Use-Case Forum	- All required use-case-specific alignment attributes populated - Explicit confirmation of, for example: - Functional equivalence - System boundary consistency - Declared unit consistency - Primary Data Share (PDS) consistency - Consistency on LCA database used - Consistency on LCA software used - PCF shared alongside supplementary documentation required for the specific decision	- Level + status is not achieved if required alignment conditions for the specific use case are missing - Comparability or claim validity breaks if assumptions differ or are insufficiently documented - Alignment requirements differ depending on the application context (e.g., tendering vs marketing claim vs certification)

Component 1. PCF Levels

Business Area	Business Decision / Application of PCF Data	Level 1	Level 2	Level 3	Level +
Supplier engagement and procurement	Identify priority product categories for decarbonization planning	Fit for purpose	Fit for purpose	Fit for purpose	
	Refine/prioritize suppliers' engagement	Fit for purpose	Fit for purpose	Fit for purpose	
	Decide whether to invite a supplier into a joint decarbonization or capability-building program	Fit for purpose	Fit for purpose	Fit for purpose	
	Cluster suppliers (services or homogeneous portfolios) by relative carbon intensity	Directional insight	Fit for purpose	Fit for purpose	
	Use PCF data as a criterion in carbon-informed tendering processes	Not fit for purpose	Directional insight	Fit for purpose	
	Decide to pay/refuse a green premium based on PCF comparison for the same material	Not fit for purpose	Not fit for purpose	Not fit for purpose	Required
Internal portfolio and intervention planning	Prioritize internal decarbonisation interventions across products or portfolios	Directional insight	Fit for purpose	Fit for purpose	
	Evaluate the credibility and potential impact of the Supplier PCF Reduction Plan	Directional insight	Fit for purpose	Fit for purpose	
	Identify specific SKUs or part numbers for decarbonization efforts	Not fit for purpose	Fit for purpose	Fit for purpose	
	Refine Scope 3.1 hotspot analysis using supplier-specific PCF data	Not fit for purpose	Fit for purpose	Fit for purpose	
Reporting and disclosure	Track and report year-on-year Scope 3.1 emissions performance, incorporating supplier PCF data	Not fit for purpose	Directional insight	Fit for purpose	
	Integrate supplier-specific PCF data into Scope 3.1 inventory calculations	Not fit for purpose	Not fit for purpose	Fit for purpose	
	Use PCF data as an input for EU Battery Regulation / EU Digital Product Passport reporting requirements	Not fit for purpose	Not fit for purpose	Fit for purpose	
	Prioritize engineering / redesign effort across SKUs or part numbers	Not fit for purpose	Directional insight	Fit for purpose	
Product and engineering	Analyze carbon performance trends across material classes	Not fit for purpose	Directional insight	Fit for purpose	
	Use supplier PCF data as an input to publishing or certifying your own product-level PCF or EPD	Not fit for purpose	Directional insight	Fit for purpose	
	Use PCF data as an informational input for marketing / launching a "lower-carbon" product variant	Not fit for purpose	Directional insight	Fit for purpose	
	Compare "Material A" from Supplier 1 and Supplier 2	Not fit for purpose	Not fit for purpose	Not fit for purpose	Required

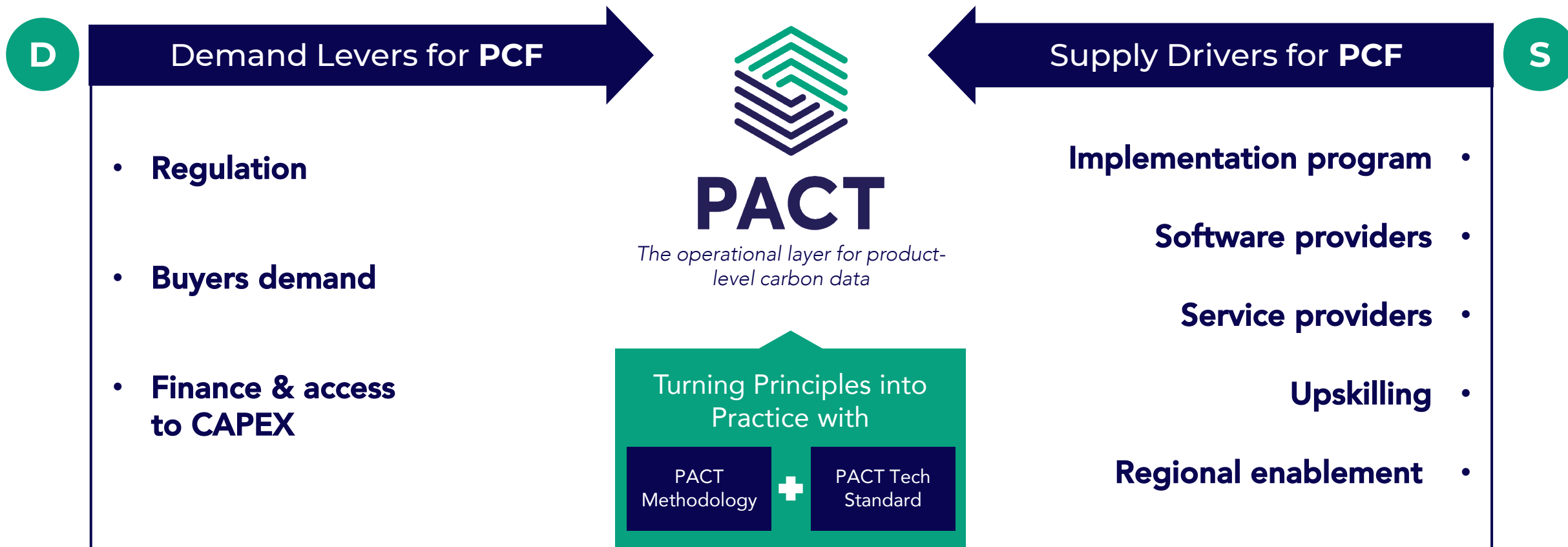
Component 2. Fit-for-Purpose Matrix



The **PCF Usability Model** introduces a framework that links data **completeness** and appropriate levels of assurance to **specific use cases** of product-level carbon data.

From principles to practice: PACT as the engine for PCF market growth

PACT is now scaling globally by
accelerating **demand** and **supply** of high-quality Product-level Carbon Data



We will now hear from..



Add your
questions
scanning the QR
code

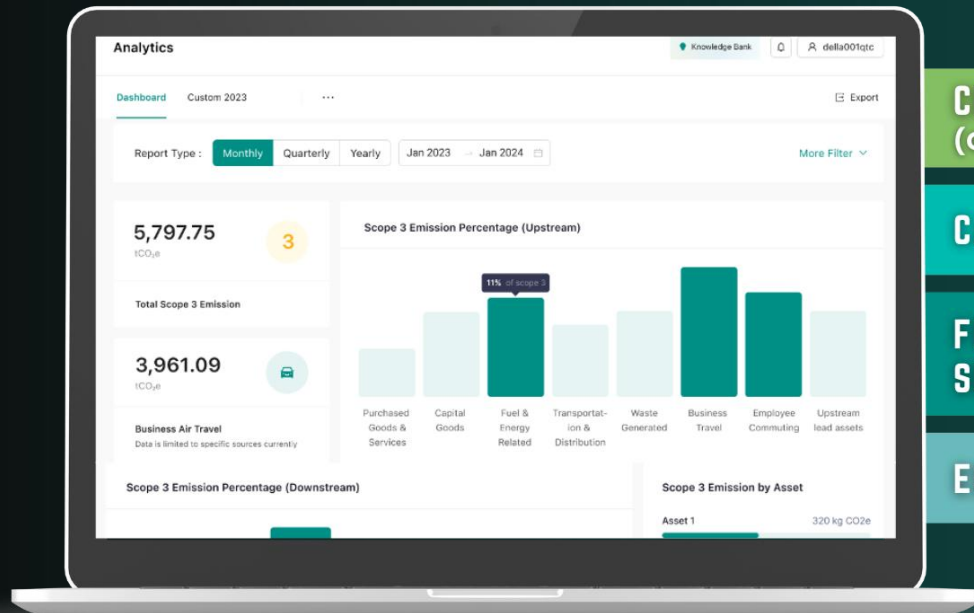


Benjamin Tan

VP Business
Development,
ESGPedia

Technology Sharing

Supporting Asia's Net Zero Landscape with PACT



CREATE ESG AND GHG REPORTS
(GRI, ISSB, ISO 14064, and more)

CALCULATE AND TRACK CARBON EMISSIONS

FULL VALUE CHAIN
SUSTAINABILITY AND ANALYTICS

ESG AND DECARBONISATION SOLUTIONS



Get in touch

Introduction to ESGpedia - Trusted By >1000 Companies Across Asia-Pacific

Real Estate & Construction



FMCG & Manufacturing



Technology, Electronics & Energy



Healthcare



Logistics & Transport



Financial Institutions



Events & Hospitality



Educational Institutions



Why Sustainability and Net-Zero Matters to Businesses

Public Sector



GreenGov.SG Targets

- **2025:** Public sector emissions expected to peak around 2025 before declining
- **2028:** Sustainability criteria included in **all** government procurement decisions,
- **2045:** Net zero emissions

Government Tenders

5-10% of evaluation points in government tenders.



Table 1: Examples of the Environmental Sustainability Criteria

Environmental Sustainability Criteria
Climate-related disclosure Tenderer has published climate-related disclosures in accordance with international standards.
Decarbonisation of construction processes Tenderer has committed to adopt practices that decarbonize the construction work processes such as electrification of construction equipment that are conventionally fuel-powered and has developed a plan to implement them for the project.
Green certifications and qualifications a) Any of the ISO certifications below: i. ISO14001: Environmental Management Systems; or ii. ISO50001: Energy Management Systems; or iii. ISO46001: Water Efficiency Management Systems b) One or more individuals certified GMAP or GMAAP and who are assigned to the project. c) Green and Gracious Builder Scheme (GGBS) Merit and above
Innovation Tenderer may propose sustainable construction practices complete with analysis on the environmental benefits and emission impact, for GPE's evaluation. Example of such practice: • Use of low-carbon materials above and beyond specifications • Use of renewable energy to support site operations • Use of low-carbon transportation options to site

Private Sector



Mandatory climate disclosures from 2025 for SGX-listed firms

Covers:

- Greenhouse gas emissions (Scope 1-3)
- Energy consumption and resource use
- Climate-related risks and opportunities
- Transition plans and targets

Commercial Tenders

Prefer suppliers who align with their **Sustainable Procurement Policies**.

Supplier screening with ESG criteria (e.g. emissions, waste, resource usage, ethical sourcing, certifications, etc.)



Growing Demand for PCF to streamline BCA Green Mark 2021 and SGBP Certification

Green Mark Ready

1. End-to-End Carbon Coverage

- Embodied (materials) + operational (energy use)
- Aligned with Whole Life Carbon (Cn)

2. Material & Hotspot Insights

- Breakdown by key materials (concrete, steel)
- Identifies carbon hotspots for reduction

Impact:

1. Higher Scores

- Secure **GoldPLUS / Platinum** ratings
- Improves WLC (Cn) scoring

2. Increased Transparency & Faster Delivery

- From months to days
- **Audit-ready, traceable data**

Aligned with BCA Green Mark 2021



The [Whole Life Carbon section](#) will score buildings on their Whole Life Carbon performance based on international calculation methodologies, upfront (embodied) carbon (i.e., carbon footprint of the construction materials), use of sustainable construction methods, and the sustainable fitting-out of buildings. The section will also evaluate the plans, and delivery of those plans to transition the building towards zero carbon emissions.

	Energy Savings				Intelligence	Health & Well-being	Whole Life Carbon	Maintainability	Resilience
	>50%	≥55%	≥60%	ZE	15 points for each sustainability section ≥10 points in a section to qualify for a badge for exemplary performance				
A. Full GM: 2021 certification									
SLE (incl. ZE, PE)			✓	✓	N/A				
Platinum SLE (incl. ZE, PE)			✓	✓	40 points				
Gold ^{PLUS} SLE (incl. ZE, PE)			✓	✓	30 points				
Platinum		✓			40 points				
Gold ^{PLUS}	✓				30 points				

Points will be awarded for products with SGBP certification. SGBP Certified Products are highly recognised under the Green Mark, allowing projects to accrue points towards a Green Mark rating.

Supporting toward SGBP Certification

Under the latest SGBP criteria revisions, steel and ready-mixed concrete products are now being evaluated based on GWP reported in accepted EPDs for certification. SGBC works with industry collaborators via its technical committees to review other product categories and when relevant, would shift towards GWP reporting via EPDs.

Case Study: PCF and EPD for Concrete & Building Material Manufacturer

Background

Integrated Precast Solutions and similar manufacturers needed to obtain **EPD** for products to meet **SGBP certification requirements** and **growing customer demand**.

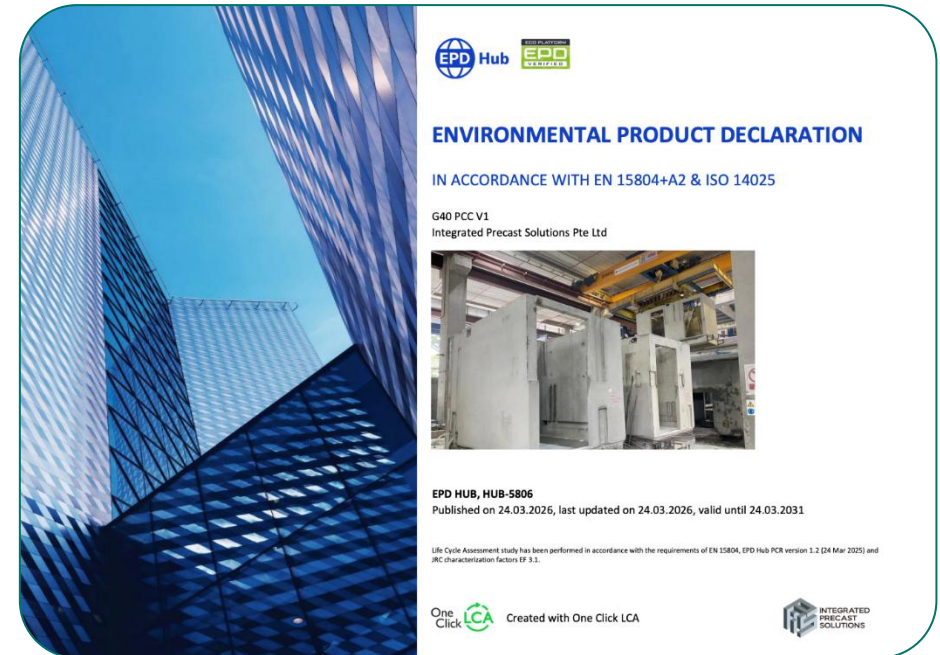
Challenges:

- EPD process is **complex and time-consuming**
- **Lack of in-house expertise** for LCA and documentation
- Need for **third-party verification** and official publication

Solution

ESGpedia supported manufacturers with **end-to-end EPD development, verification, and registry submission**

- ✓ Successfully enabled the **development and publication of Product EPD reports**, including **third-party verification**
- ✓ Simplified the EPD process through **hands-on guidance**, **reducing client effort**.



Download Sample EPD Report

Scan QR or click below:



[View Report on EPD Hub](#)



ESGpedia is a PACT Conformant Solution



ESGpedia



DEMO - Connecting into PACT Network

Benefits

- **Allows product carbon footprint data to be seamlessly uploaded into PACT**
- **Amplifying visibility and awareness of the product, that the product is aligned with PACT Standard**
- **Providing transparency and confidence to end-customers with the need for lower-carbon products as part of their net-zero plans, and for green certifications**

Demo Video



We will now hear from..



CK Chung

Founder & Managing
Director, Smart Tradzt



Add your
questions
scanning the QR
code

Carbon Reporting Was Never The End Goal: Turning Product Carbon Data Into Revenue, Procurement and Financing Outcomes

How interoperable carbon data enables Scope 3 reduction,
premium capture and transition financing in the built environment.



Trusted Data



Commercial
Execution



Real Impact



ABOUT SMART TRADZT

Climate Tech Software / Consulting / Training



OUR RECOGNITIONS



- United Overseas Bank's Greentech Accelerator for categories
 - Green Procurement
 - Product Carbon Footprint



- Climate Finance Innovation Lab (CFIL)
- Next Frontier Building / Real Estate Decarbonisation
- Organised by JC3 (Malaysia Central Bank, Security Commission)
- Supported by UNDP



- WBCSD (World Business Council for Sustainable Development) PACT conformant software



Headquarter
in Singapore



ISO27001 Certified

SAMPLE CLIENTS



The Market Problem

Carbon Reporting Was Never The End Goal

Current market failure:



Developers want lower embodied carbon buildings



Steel producers invest in decarbonisation but struggle to monetize premiums



Banks want transition finance opportunities but lack trusted data



Buyers don't trust carbon claims



Supply chains are fragmented



Result:

Carbon reporting remains disconnected from business execution.

“

The real challenge is not carbon accounting. It is converting carbon data into **commercial action**.



Trusted Data



Commercial Execution

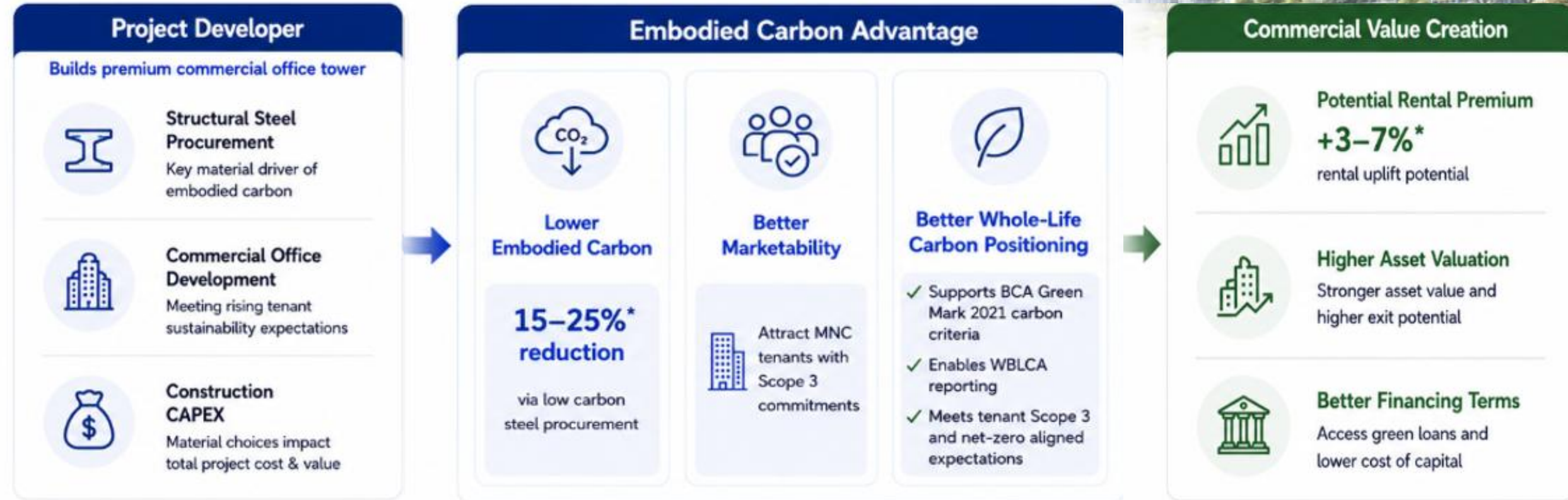


Real Impact



Why Developers Will Pay for Lower-Carbon Materials

Embodied carbon reduction is increasingly linked to asset value, tenant demand and financing advantages.



Carbon reduction must improve asset economics— not just sustainability reporting.

* Illustrative estimates based on green building market studies; actual outcomes vary by market.
Source: BCA Green Mark 2021 (Whole Life Carbon Section), USGBC (LEED v4.1 Materials & Resources), World Green Building Council, JLL, MSCI, BCG, McKinsey – industry research and illustrative estimates



Traditional Procurement Cannot Deliver Deep Carbon Reduction

Carbon outcomes are determined upstream.
Procurement models must evolve to shift influence to the source.



You cannot decarbonise what you do not directly procure.



Direct Influence
Carbon outcomes shaped at the source



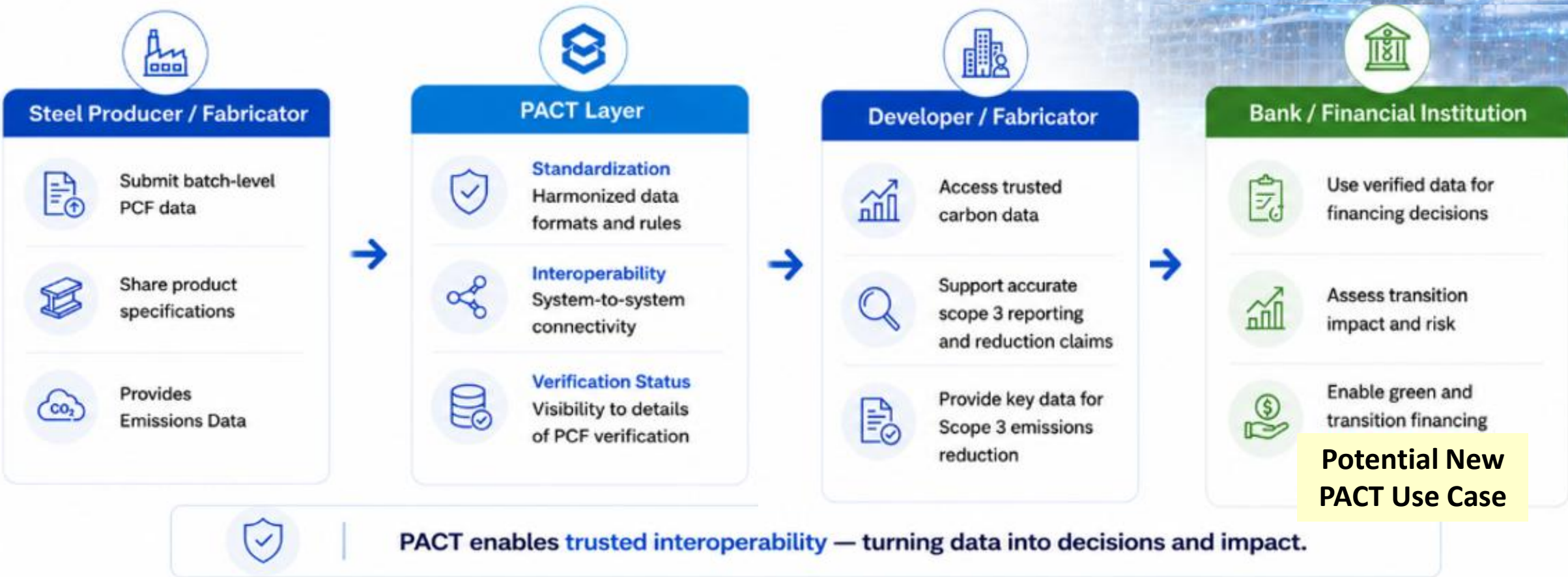
Better Outcomes
Lower carbon, better economics



Stronger Partnerships
Aligned incentives across the value chain

How PACT Enables Trusted Carbon Exchange

PACT creates a common language and trusted layer for carbon data across the steel value chain.



Potential New
PACT Use Case

Trusted Data

Verified, validated and auditable

Seamless Exchange

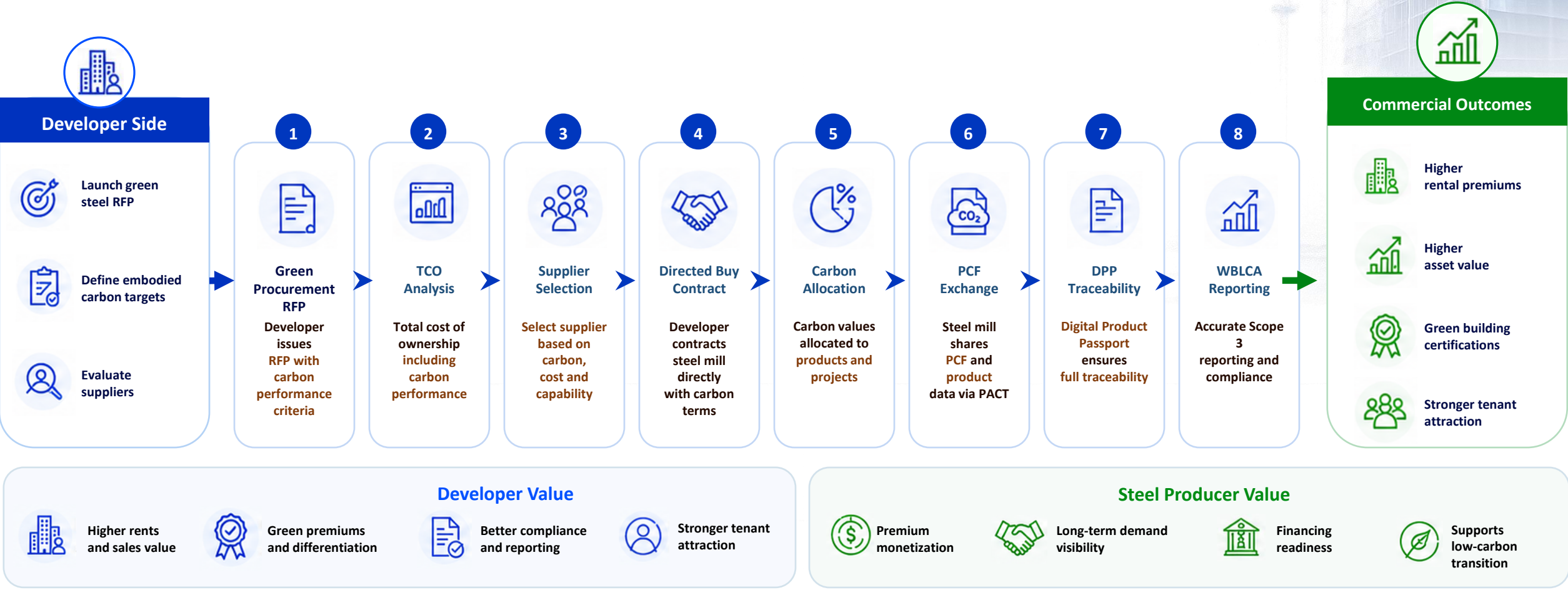
Across systems and organizations

Actionable Impact

Better decisions, lower emissions

From Carbon Data to Commercial Execution

PACT supports End to End Scope 3 reduction execution - from PCF exchange to Commercial Value Realisation.
Carbon Transparency becomes commercial advantage.



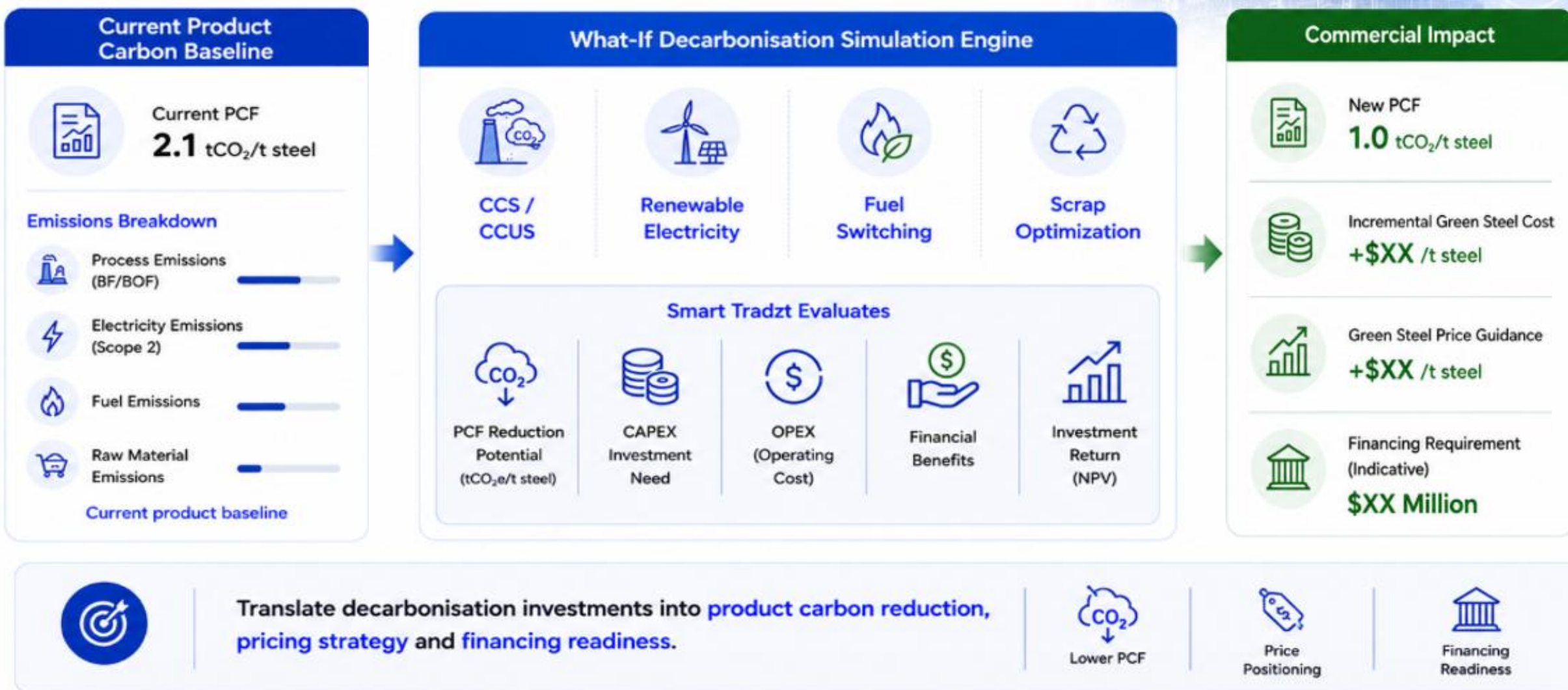
This is where carbon reporting becomes economic execution.

Launch Video 1

From Carbon Disclosure to Decarbonisation Planning

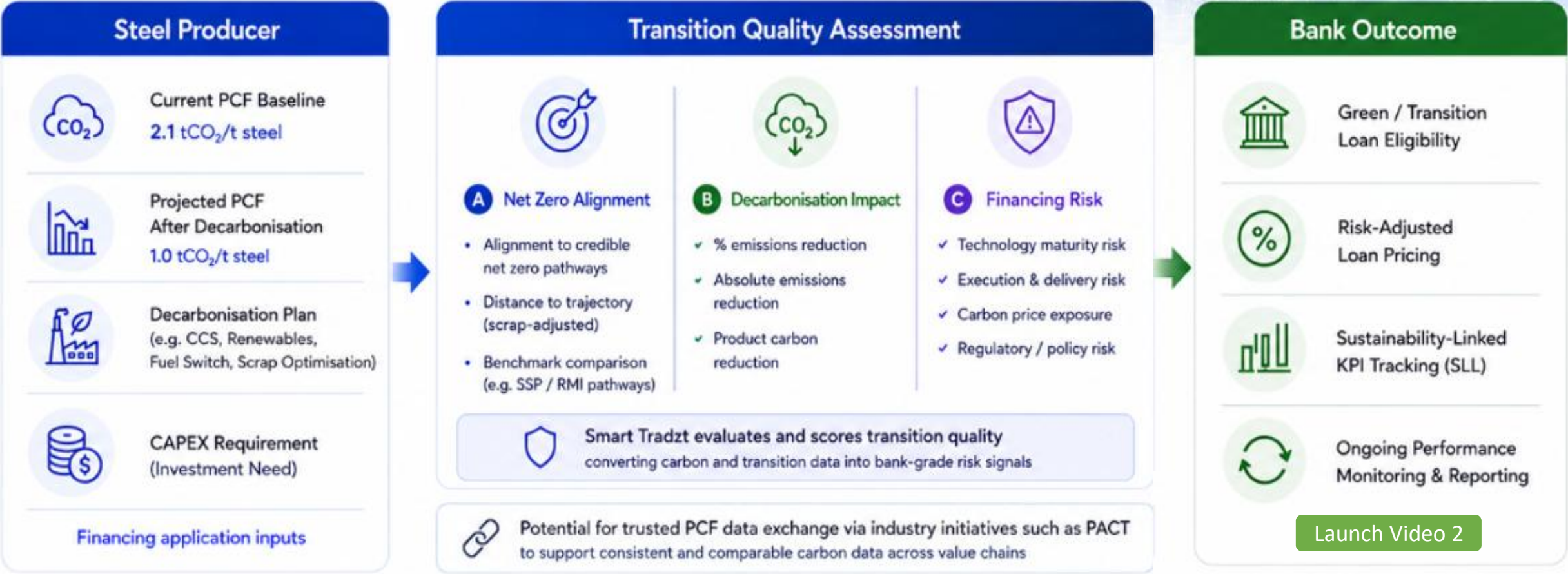
Smart Tradzt helps steel producers simulate decarbonisation pathways, quantify PCF reduction, and assess commercial viability.

CO₂



From Decarbonisation Planning to Bankable Financing

Smart Tradzt turns product carbon data and transition plans into bankable financing decisions for steel producers.



Our platform enables banks to **finance credible steel transition projects** and **accelerate embodied carbon reduction at scale.**

Scalable Steel Decarbonisation

Lower Embodied Carbon Buildings

Faster Scope 3 Reduction

Carbon Reporting Was Never The End Goal

The future belongs to companies that turn carbon data into commercial execution infrastructure.



Carbon data is becoming **economic infrastructure**.

Smart Tradzt connects carbon data, commercial workflows and financing to accelerate **embodied carbon reduction—at scale**.



Enable low-carbon
procurement
with confidence



Empower steel producers
to decarbonise
with clarity



Unlock green financing
for credible transition
at scale



Drive real-world
impact across
the built environment

From footprint to impact. From data to value. From ambition to action.

Capital is available. What's missing is data reliable enough to act on.

Three decisions that verified product carbon data transforms.

COMERCIAL BANKS

Green & sustainability-linked lending

WITHOUT VERIFIED DATA

Supplier claims low carbon. No way to verify. Bank defaults to standard rate, or walks away.



WITH VERIFIED DATA

PACT-based Product Carbon Data confirms performance. Bank offers a differentiated green rate. Supplier is rewarded for real progress.

DEV. BANKS & INVESTORS

Transition bonds & blended finance

WITHOUT VERIFIED DATA

Bond labelled 'transition'. No product-level data behind the claim. Greenwashing risk. Investors hesitate.



WITH VERIFIED DATA

PACT data provides a credible emissions baseline and trajectory. The bond is defensible. Capital flows at scale.

ALL FINANCE INSTITUTIONS

Supply chain finance & risk pricing

WITHOUT VERIFIED DATA

Supply chain carbon exposure is opaque. Banks can't price transition risk accurately. Regulatory pressure is rising.



WITH VERIFIED DATA

Verified supplier Product Carbon Data maps carbon exposure across the portfolio. Risk becomes visible, priceable, manageable.

Why act now?

The case for financial organizations to build on reliable carbon data infrastructure today.

OPPORTUNITY

The fastest-growing lending product needs better data

Verified PCF data is the missing KPI. Without it, SLL pricing defaults to assumption and opportunity is lost.

**\$1.7
trillion**
global SLL market

RISK

Transition risk exposure is invisible without supply chain data

Supply chain carbon exposure is the hardest gap to fill, and the one regulators are watching most closely.

50%+
of Scope 3 sits in
supply chains

ADVANTAGE

First-mover advantage is compounding

Early access to supplier carbon data is a portfolio advantage that gets harder to build as the market matures. Infrastructure is harder to replicate than capital.

2026
CBAM fully
enforced. supplier
data race begins

IMPACT

Real, measurable impact on supply chain decarbonisation

Capital directed to verified low-carbon suppliers creates measurable, reportable Scope 3 reduction. Credible enough to put in your net-zero report.

€12bn
annual CBAM cost
driving supplier
data urgency

CBAM turns carbon into a cost factor

What is changing?

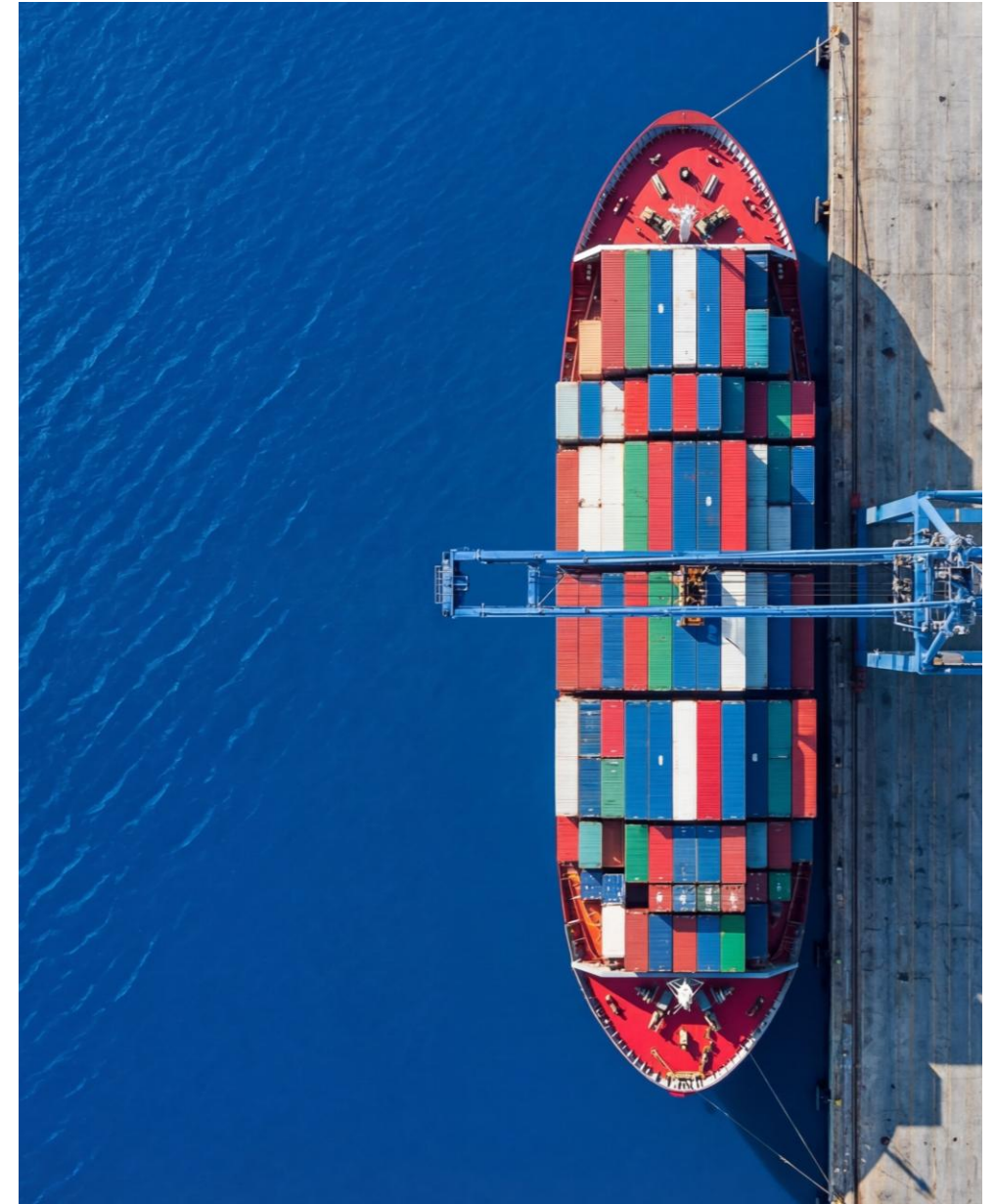
- CBAM moves from reporting to **real financial exposure**: embedded emissions X EUA price
- For steel/aluminium/cement, CBAM becomes a per-tonne cost component like shipping/duties.

Hard evidence

- With EUA prices in the €80- €90/tCO₂ range, high carbon steel faces €80- €120/ton added cost
- CBAM charges for hot-rolled coil (HRC) can reach faces €80- €174/ton

So what? Implications for business

Default values amplify cost/dispute risk, credible. Auditable, comparable carbon data becomes a key competitive edge.



Verified emissions data can cut CBAM costs by ~60%

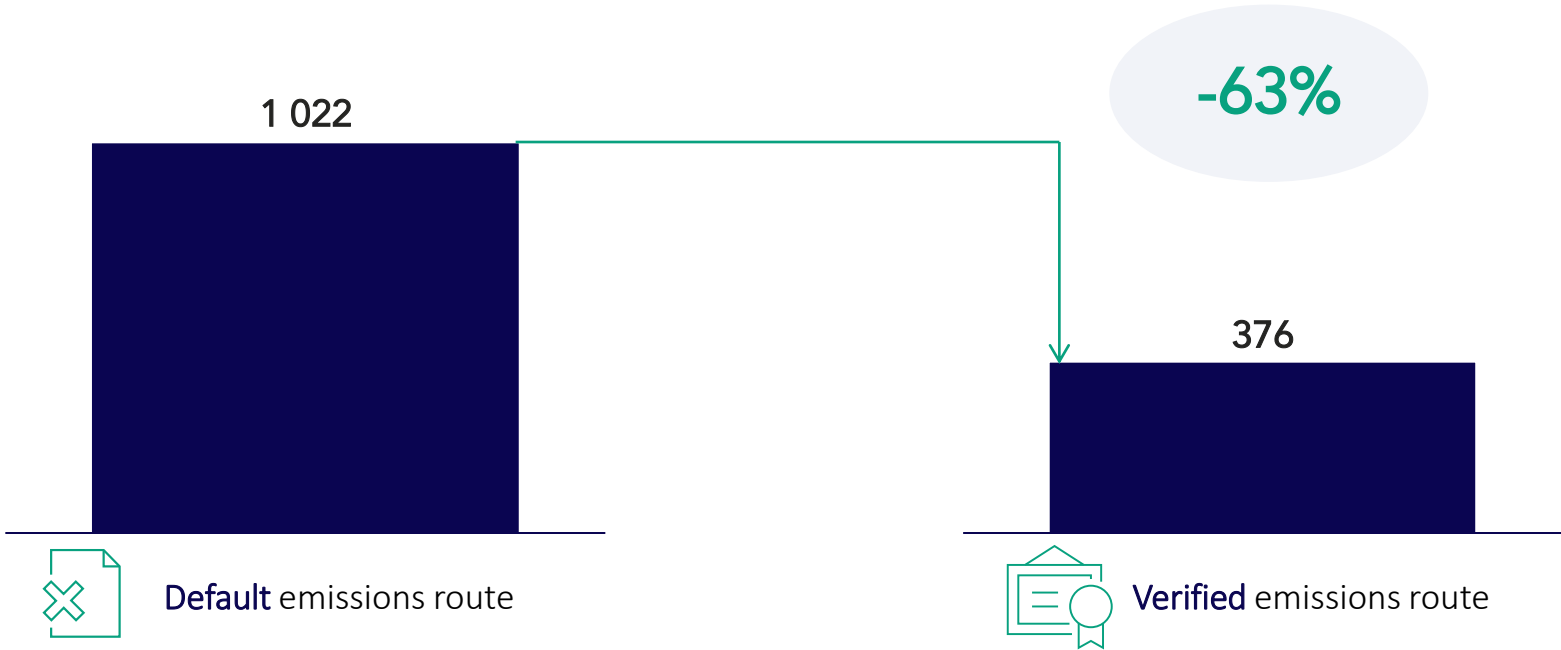
Verified emissions data also enables supply chain optimization and other actions to reduce the liability

Case example

Carbon cost of CBAM certificates, M€

Context:

- EU importer purchases 3,000 tons of steel from China in 2026 (CN code 7228 40)
- CBAM price: € 88/tCO₂
- Hypothetical recognized carbon price in China: € 15/tCO₂



1. [EC Implementation Act](#) – default value including 10% surcharge; 2. Rounded estimate based on public [study](#) estimating value of 2.33 tCO₂/t of steel production; 3. [EC Implementation Act](#); 4. Benchmark calculation for verified emissions needs to be based on the entire value chain aggregation; the value assumed here is based on the proportional reduction seen for the embedded emissions

One carbon data foundation, multiple outputs

CBAM as proof point



Methodology guidance

Dimension-by-dimension PCF-CBAM analysis: what data is needed, where gaps are structural, what bridging requires



Data architecture & implementation

Define the portable carbon data package that serves multiple regulatory endpoints from one collection effort



Pilot & validate with members

Test approach with working group companies, buyers, solution providers, and suppliers, to validate feasibility

Your speakers today



Naama Avni-Kadosh

Director, PACT



Benjamin Tan

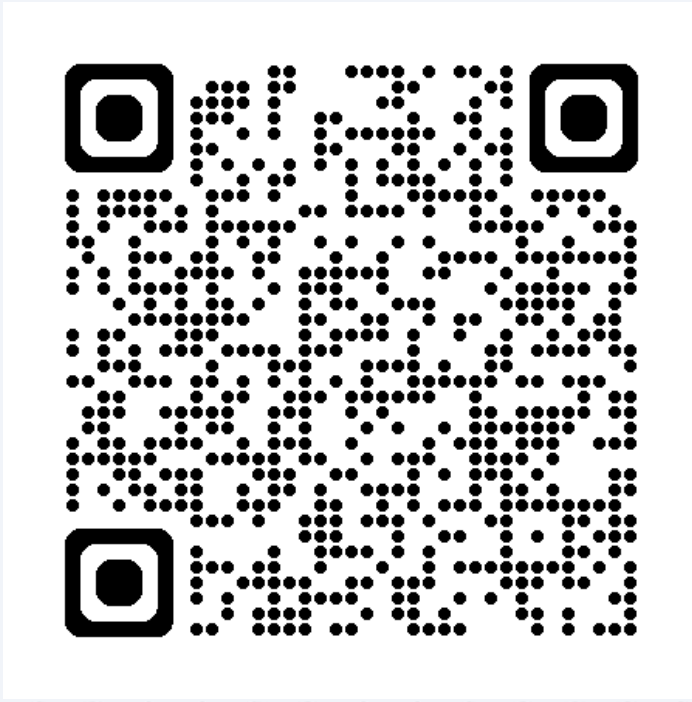
VP Business Development,
ESGPedia



CK Chung

Founder & Managing
Director, Smart Tradzt

Your pocket guide to *PACT*



*Download the
PACT Cheat Sheet
Everything your team
needs, in one place*

Turn your sustainability data into a business asset with PACT

Use PACT Infrastructure

- Adopt PACT methodology / PACT aligned industry specific methodology
- WBCSD Implementation program
- PACT conformant IT solution
- PACT aligned consultancy

Design PACT Infrastructure

- Join the CBAM pilots and working groups
- Test PACT Usability Model
- Design next data packages in PACTs Use case forum
- Pilot data exchange with PACT Sand Box environment

Contact us for resources, support, and inquiries: pact@wbcsd.org

APAC Upcoming Key Events

*Share your feedback
to improve our future
events:*



Thank you

Contact us at apac@wbcSD.org if you would like to know more about this event or WBCSD.

